ITNOW: the nowcasting model

Methodological note 17 October 2026

We nowcast Italian quarterly GDP in chain-linked terms using industrial production (IP), EUROCOIN and two static principal-component factors. The factors summarise our monthly panel of 19 indicators, covering activity, prices, surveys and financial conditions. We update the same equation each month, as new observations become available.

The equation. Let Y_q be GDP and $y_q = \log(Y_q/Y_{q-1})$ its quarterly log growth. If t_q is the last month of quarter q, we estimate

$$y_q = \alpha + \beta \log\left(\frac{\overline{IP}_q}{\overline{IP}_{q-1}}\right) + \gamma \overline{EC}_q + \delta_1 s_{1q} + \delta_2 s_{2q} + u_q,$$

$$s_{jq} = f_{j,t_q} + 2f_{j,t_q-1} + 3f_{j,t_q-2} + 2f_{j,t_q-3} + f_{j,t_q-4}.$$

Bars denote arithmetic averages of the three monthly observations. EUROCOIN enters in levels. Each factor uses five months with fixed weights $[1, 2, 3, 2, 1]$. For Q3, for example, these weight September, August, July, June and May. They are a fixed aggregation rule; we do not estimate them.

Estimation. At each forecast month, we transform and standardise the available panel using its expanding history from May 1996. Most indicators enter as log changes; we use locally seasonally adjusted changes for HICP and tourism, and a change in the inverse-hyperbolic-sine transform for trade balance. ITACOIN and EUROCOIN enter the panel in levels. Already adjusted source series retain their source adjustment.

For PCA, we use indicators observed in at least 20% of this history. We fill interior gaps by splines and remaining gaps by regressions on the five most correlated auxiliary series. Available GDP also helps this imputation. We extract two principal components from the completed monthly indicators and scale them by their component standard deviations. We rebuild their full histories at each forecast month and exclude the first four monthly observations from the bridge regressors. We then estimate the bridge by ordinary least squares, with an intercept, using all complete, released quarters and at least 12 observations. The GDP response remains in log-growth units.

Missing months and timing. We use fixed publication delays: two months for IP and one month after quarter-end for GDP; EUROCOIN has no additional monthly delay. The other panel indicators follow their recorded publication lags. We project unreleased IP levels, EUROCOIN and future factor values at their last available values. Actual gaps in the IP or EUROCOIN history remain missing. There are no COVID sign restrictions or caps.

Reporting and evaluation. We report ordinary QoQ growth as $100[\exp(\widehat{y}_q) - 1]$ and recover the GDP level as $Y_{q-1}\exp(\widehat{y}_q)$. The 90% prediction interval uses the linearly interpolated 5th and 95th percentiles of earlier forecast errors from the same month within the quarter, once 40 released quarterly errors are available. Our historical exercise uses a revised data snapshot and fixed delays, so it is pseudo real time. We selected this specification after comparing historical performance; online evaluation will assess it going forward. The separate GDP note explains YoY growth and annual carry-over.